Name: Hashir Ahmed Buriro **Python Developer Task 1** **Date: 28/4/2026**

PYTHON DEVELOPER INTERNSHIP PROGRAM

Hasnain Karimain Educational Academy

Software House & Training Center

Registered with SECP & PSEB (Ministry of IT, Government of Pakistan)

Shift: 1st Tasks / Projects Assigned Days: Monday to Wednesday

Task 1: Advanced CLI Contact Management System with Search & Filters

🎯 Objective

Develop a professional **Command-Line Contact Management System** that allows users to store, manage, search, and filter contacts efficiently. The system should simulate a real-world mini-CRM (Customer Relationship Management tool).

🛠️ Task Description

Create a Python-based CLI application where users can:

- Add new contacts
- View all contacts
- Search contacts by name, phone, or email
- Apply filters (e.g., by city or company)
- Update existing contact details
- Delete contacts
- Save and load data from a file (JSON/CSV)

The system should be interactive, user-friendly, and modular.

⚙️ Functional Requirements

Contact Fields:

Each contact must include:

- Full Name
- Phone Number
- Email Address
- City
- Company

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Core Features:

1. **Add Contact**
   - Validate input (no empty fields, valid email/phone)
2. **View Contacts**
   - Display in formatted table style
3. **Search Contacts**
   - By Name (partial match)
   - By Phone
   - By Email
4. **Filter Contacts**
   - By City
   - By Company
5. **Update Contact**
   - Edit specific fields
6. **Delete Contact**
   - Delete by ID or Name
7. **Data Persistence**
   - Save contacts in JSON or CSV
   - Load data automatically on program start

Advanced Requirements (Must Have)

- Use **functions** for each feature
- Implement **menu-driven CLI interface**
- Use **exception handling** for errors
- Implement **search optimization (case-insensitive)**
- Add **unique ID for each contact**
- Use **modular code structure (separate functions/files optional)**

Bonus Features (Optional but Recommended)

- Export contacts to CSV
- Import contacts from file
- Add favorite/starred contacts
- Pagination for large data
- Sorting (A-Z, Recently Added)

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Deliverables

Intern must submit:

1. ✅ Source Code (ZIP file)
2. ✅ GitHub Repository
3. ✅ README.md (features + how to run)
4. ✅ Sample Data File (JSON/CSV)
5. ✅ Demo Video (1–2 minutes)

⏰ Deadline

24 Hours

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Project Report

Objective

The objective of this project is to develop a **Command-Line Contact Management System (CRM-like application)** using Python. The system allows users to efficiently store, manage, search, filter, update, and delete contact information with data persistence.

Project Description

This project is a **menu-driven CLI application** built in Python. It simulates a basic Customer Relationship Management (CRM) system where users can manage contacts in a structured and efficient way.

All contact data is stored in a **JSON file**, ensuring that data is saved permanently even after the program closes.

CODE:

Main.py

```
import json
import os
import re
from datetime import datetime

FILE_NAME = "contacts.json"


# =========================
# LOAD DATA
# =========================
def load_contacts():
    if os.path.exists(FILE_NAME):
        try:
            with open(FILE_NAME, "r") as file:
                return json.load(file)
        except:
            return []
    return []


# =========================
# SAVE DATA
```

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```python
# ==========================
def save_contacts(contacts):
    with open(FILE_NAME, "w") as file:
        json.dump(contacts, file, indent=4)


# ==========================
# ID GENERATOR
# ==========================
def generate_id(contacts):
    if not contacts:
        return 1
    return max(c["id"] for c in contacts) + 1


# ==========================
# VALIDATIONS
# ==========================
def validate_email(email):
    pattern = r'^[\w\.-]+@[\w\.-]+\.\w+$'
    return re.match(pattern, email)


def validate_phone(phone):
    pattern = r'^\+?\d{10,15}$'
    return re.match(pattern, phone)


# ==========================
# ADD CONTACT
# ==========================
def add_contact(contacts):
    print("\n=== ADD CONTACT ===")

    name = input("Name: ").strip()
    phone = input("Phone: ").strip()
    email = input("Email: ").strip()
    city = input("City: ").strip()
    company = input("Company: ").strip()

    if not all([name, phone, email, city, company]):
        print("❌ All fields are required.")
        return

    if not validate_email(email):
        print("❌ Invalid email.")
```

```
        return

    if not validate_phone(phone):
        print("❌ Invalid phone number.")
        return

    contact = {
        "id": generate_id(contacts),
        "name": name,
        "phone": phone,
        "email": email,
        "city": city,
        "company": company,
        "favorite": False,
        "created_at": str(datetime.now())
    }

    contacts.append(contact)
    save_contacts(contacts)

    print("✅ Contact added successfully!")


# ========================
# VIEW CONTACTS (FIXED OUTPUT)
# ========================
def view_contacts(contacts):
    if not contacts:
        print("\n⚠️ No contacts found.")
        return

    print("\n================ CONTACT LIST ================\n")

    header = f"{'ID':<5} {'Name':<20} {'Phone':<15} {'Email':<25} {'City':<15}
{'Company':<15}"
    print(header)
    print("-" * len(header))

    for c in contacts:
        print(
            f"{c['id']:<5} "
            f"{c['name']:<20} "
            f"{c['phone']:<15} "
            f"{c['email']:<25} "
            f"{c['city']:<15} "
```

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```
            f"{c['company']:<15}"
        )

    print("\n==============================================\n")


# =========================
# SEARCH CONTACTS
# =========================
def search_contacts(contacts):
    keyword = input("\nSearch (name/phone/email): ").lower()

    result = [
        c for c in contacts
        if keyword in c["name"].lower()
        or keyword in c["phone"]
        or keyword in c["email"].lower()
    ]

    if not result:
        print("❌ No match found.")
        return

    view_contacts(result)


# =========================
# FILTER CONTACTS
# =========================
def filter_contacts(contacts):
    print("\n1. Filter by City")
    print("2. Filter by Company")

    choice = input("Choose: ")

    if choice == "1":
        city = input("Enter city: ").lower()
        result = [c for c in contacts if c["city"].lower() == city]

    elif choice == "2":
        company = input("Enter company: ").lower()
        result = [c for c in contacts if c["company"].lower() == company]

    else:
        print("❌ Invalid choice.")
```

```python
        return

    if not result:
        print("❌ No contacts found.")
        return

    view_contacts(result)


# ========================
# UPDATE CONTACT
# ========================
def update_contact(contacts):
    try:
        cid = int(input("\nEnter Contact ID: "))
    except:
        print("❌ Invalid ID")
        return

    for c in contacts:
        if c["id"] == cid:

            print("Leave blank to keep old value\n")

            name = input(f"Name ({c['name']}): ").strip()
            phone = input(f"Phone ({c['phone']}): ").strip()
            email = input(f"Email ({c['email']}): ").strip()
            city = input(f"City ({c['city']}): ").strip()
            company = input(f"Company ({c['company']}): ").strip()

            if name:
                c["name"] = name
            if phone:
                if validate_phone(phone):
                    c["phone"] = phone
            if email:
                if validate_email(email):
                    c["email"] = email
            if city:
                c["city"] = city
            if company:
                c["company"] = company

            save_contacts(contacts)
            print("✅ Updated successfully!")
```

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```python
            return

    print("❌ Contact not found")


# ========================
# DELETE CONTACT
# ========================
def delete_contact(contacts):
    try:
        cid = int(input("\nEnter Contact ID: "))
    except:
        print("❌ Invalid ID")
        return

    for c in contacts:
        if c["id"] == cid:
            contacts.remove(c)
            save_contacts(contacts)
            print("✅ Deleted successfully!")
            return

    print("❌ Not found")


# ========================
# SORT CONTACTS
# ========================
def sort_contacts(contacts):
    contacts.sort(key=lambda x: x["name"].lower())
    print("✅ Sorted A-Z")
    view_contacts(contacts)


# ========================
# FAVORITE CONTACT
# ========================
def favorite_contact(contacts):
    try:
        cid = int(input("\nEnter Contact ID: "))
    except:
        print("❌ Invalid ID")
        return

    for c in contacts:
```

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```
        if c["id"] == cid:
            c["favorite"] = True
            save_contacts(contacts)
            print("☆ Marked as favorite")
            return

    print("✕ Not found")


# ========================
# MAIN MENU
# ========================
def main():
    contacts = load_contacts()

    while True:
        print("\n===== CONTACT MANAGEMENT SYSTEM =====")
        print("1. Add Contact")
        print("2. View Contacts")
        print("3. Search Contacts")
        print("4. Filter Contacts")
        print("5. Update Contact")
        print("6. Delete Contact")
        print("7. Sort Contacts")
        print("8. Favorite Contact")
        print("9. Exit")

        choice = input("Enter choice: ")

        if choice == "1":
            add_contact(contacts)
        elif choice == "2":
            view_contacts(contacts)
        elif choice == "3":
            search_contacts(contacts)
        elif choice == "4":
            filter_contacts(contacts)
        elif choice == "5":
            update_contact(contacts)
        elif choice == "6":
            delete_contact(contacts)
        elif choice == "7":
            sort_contacts(contacts)
        elif choice == "8":
            favorite_contact(contacts)
```

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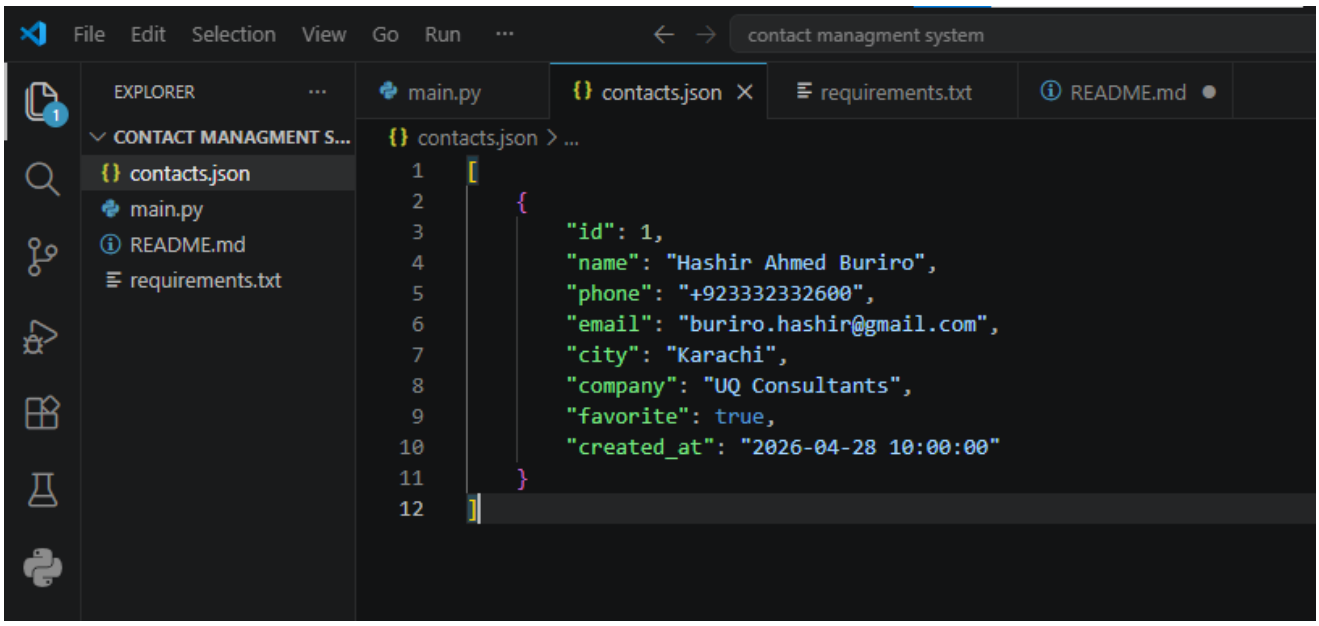
```
        elif choice == "9":
            print("👋 Exiting...")
            break
        else:
            print("❌ Invalid choice")


# ========================
# RUN PROGRAM
# ========================
if __name__ == "__main__":
    main()
```

Contact.json



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OUTPUT:

```
Edit  Selection  View  Go  Run  ···   contact managment system

EXPLORER ···
CONTACT MANAGMENT S...
 contacts.json
 main.py
 README.md
 requirements.txt

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Hasnain Karimain (HK) Educational Academy, Software House & Training Center\contact ma

===== CONTACT MANAGEMENT SYSTEM =====
1. Add Contact
2. View Contacts
3. Search Contacts
4. Filter Contacts
5. Update Contact
6. Delete Contact
7. Sort Contacts
8. Favorite Contact
9. Exit
Enter choice: 1

=== ADD CONTACT ===
Name: Ahmed
Phone: 03332332445
Email: ahmed@gmail.com
City: islamabad
Company: abc software
✅ Contact added successfully!

===== CONTACT MANAGEMENT SYSTEM =====
1. Add Contact
2. View Contacts
3. Search Contacts
4. Filter Contacts
5. Update Contact
6. Delete Contact
7. Sort Contacts
8. Favorite Contact
9. Exit
Enter choice: 1
```

```
NTACT MANAGMENT S...
ontacts.json
ain.py
EADME.md
equirements.txt

Enter choice: 1

=== ADD CONTACT ===
Name: Buriro
Phone: 03332223344
Email: buriro@gmail.com
City: gudu
Company: uqconsulatnts
✅ Contact added successfully!

===== CONTACT MANAGEMENT SYSTEM =====
1. Add Contact
2. View Contacts
3. Search Contacts
4. Filter Contacts
5. Update Contact
6. Delete Contact
7. Sort Contacts
8. Favorite Contact
9. Exit
Enter choice: 1

=== ADD CONTACT ===
Name: Hashir Buriro
Phone: 03332332600
Email: buriro.hashir@uqconsultants.com
City: karachi
Company: UQ Consltants
✅ Contact added successfully!

===== CONTACT MANAGEMENT SYSTEM =====
1. Add Contact
2. View Contacts
3. Search Contacts
4. Filter Contacts
5. Update Contact
6. Delete Contact

TLINE
MELINE
0
```

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```
GMENT S...
  6. Delete Contact
  7. Sort Contacts
  8. Favorite Contact
  9. Exit
  Enter choice: 2
txt

  =============== CONTACT LIST ===============

  ID    Name                Phone          Email                    City           Company
  ----------------------------------------------------------------------------------------------------
  1     Hashir Ahmed Buriro +923332332600  buriro.hashir@gmail.com  Karachi        UQ Consultants
  2     Ahmed               03332332445    ahmed@gmail.com          islamabad      abc software
  3     Buriro              03332223344    buriro@gmail.com         gudu           uqconsulatnts
  4     Hashir Buriro       03332332600    buriro.hashir@uqconsultants.com karachi        UQ Consltants

  ============================================


  ===== CONTACT MANAGEMENT SYSTEM =====
  1. Add Contact
  2. View Contacts
  3. Search Contacts
  4. Filter Contacts
  5. Update Contact
  6. Delete Contact
  7. Sort Contacts
  8. Favorite Contact
  9. Exit
  Enter choice: 4

  1. Filter by City
  2. Filter by Company
  Choose: 1
  Enter city: karachi

  =============== CONTACT LIST ===============
```

```
=============== CONTACT LIST ===============

ID    Name                Phone          Email                    City           Company
----------------------------------------------------------------------------------------------------
1     Hashir Ahmed Buriro +923332332600  buriro.hashir@gmail.com  Karachi        UQ Consultants
4     Hashir Buriro       03332332600    buriro.hashir@uqconsultants.com karachi        UQ Consltants

============================================


===== CONTACT MANAGEMENT SYSTEM =====
1. Add Contact
2. View Contacts
3. Search Contacts
4. Filter Contacts
5. Update Contact
6. Delete Contact
7. Sort Contacts
8. Favorite Contact
9. Exit
Enter choice: 5

Enter Contact ID: 4
Leave blank to keep old value

Name (Hashir Buriro):
Phone (03332332600):
Email (buriro.hashir@uqconsultants.com): hashir@gmail.com
City (karachi):
Company (UQ Consltants):
✅ Updated successfully!
```

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```
✅ Updated successfully!

===== CONTACT MANAGEMENT SYSTEM =====
1. Add Contact
2. View Contacts
3. Search Contacts
4. Filter Contacts
5. Update Contact
6. Delete Contact
7. Sort Contacts
8. Favorite Contact
9. Exit
Enter choice: 2

================ CONTACT LIST ================

ID    Name                Phone          Email                   City          Company
-----------------------------------------------------------------------------------------------
1     Hashir Ahmed Buriro +923332332600  buriro.hashir@gmail.com Karachi       UQ Consultants
2     Ahmed               03332332445    ahmed@gmail.com         islamabad     abc software
3     Buriro              03332223344    buriro@gmail.com        gudu          uqconsulatnts
4     Hashir Buriro       03332332600    hashir@gmail.com        karachi       UQ Consltants

=============================================


===== CONTACT MANAGEMENT SYSTEM =====
1. Add Contact
2. View Contacts
3. Search Contacts
4. Filter Contacts
5. Update Contact
6. Delete Contact
7. Sort Contacts
8. Favorite Contact
9. Exit
```

```
9. Exit
Enter choice: 6

Enter Contact ID: 3
✅ Deleted successfully!

===== CONTACT MANAGEMENT SYSTEM =====
1. Add Contact
2. View Contacts
3. Search Contacts
4. Filter Contacts
5. Update Contact
6. Delete Contact
7. Sort Contacts
8. Favorite Contact
9. Exit
Enter choice: 7
✅ Sorted A-Z

================ CONTACT LIST ================

ID    Name                Phone          Email                   City          Company
-----------------------------------------------------------------------------------------------
2     Ahmed               03332332445    ahmed@gmail.com         islamabad     abc software
1     Hashir Ahmed Buriro +923332332600  buriro.hashir@gmail.com Karachi       UQ Consultants
4     Hashir Buriro       03332332600    hashir@gmail.com        karachi       UQ Consltants

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===== CONTACT MANAGEMENT SYSTEM =====
1. Add Contact
2. View Contacts
3. Search Contacts
4. Filter Contacts
5. Update Contact
6. Delete Contact
```

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```
✅ Sorted A-Z

================ CONTACT LIST ================

ID    Name                Phone           Email                    City            Company
-----------------------------------------------------------------------------------------------
2     Ahmed               03332332445     ahmed@gmail.com          islamabad       abc software
1     Hashir Ahmed Buriro +923332332600   buriro.hashir@gmail.com  Karachi         UQ Consultants
4     Hashir Buriro       03332332600     hashir@gmail.com         karachi         UQ Consltants

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===== CONTACT MANAGEMENT SYSTEM =====
1. Add Contact
2. View Contacts
3. Search Contacts
4. Filter Contacts
5. Update Contact
6. Delete Contact
7. Sort Contacts
8. Favorite Contact
9. Exit
Enter choice: 8

Enter Contact ID: 4
★ Marked as favorite
```

```
Enter choice: 2

================ CONTACT LIST ================

ID    Name                Phone           Email                    City            Company
-----------------------------------------------------------------------------------------------
2     Ahmed               03332332445     ahmed@gmail.com          islamabad       abc software
1     Hashir Ahmed Buriro +923332332600   buriro.hashir@gmail.com  Karachi         UQ Consultants
4     Hashir Buriro       03332332600     hashir@gmail.com         karachi         UQ Consltants

=============================================


===== CONTACT MANAGEMENT SYSTEM =====
1. Add Contact
2. View Contacts
3. Search Contacts
4. Filter Contacts
5. Update Contact
6. Delete Contact
7. Sort Contacts
8. Favorite Contact
9. Exit
Enter choice: 3

Search (name/phone/email): hashir

================ CONTACT LIST ================

ID    Name                Phone           Email                    City            Company
-----------------------------------------------------------------------------------------------
1     Hashir Ahmed Buriro +923332332600   buriro.hashir@gmail.com  Karachi         UQ Consultants
4     Hashir Buriro       03332332600     hashir@gmail.com         karachi         UQ Consltants

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```

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```
===== CONTACT MANAGEMENT SYSTEM =====
1. Add Contact
2. View Contacts
3. Search Contacts
4. Filter Contacts
5. Update Contact
6. Delete Contact
7. Sort Contacts
8. Favorite Contact
9. Exit
Enter choice: 9
👋 Exiting...
PS E:\Hasnain Karimain (HK) Educational Academy, Software House & Training Center\contact managment system> ^C
PS E:\Hasnain Karimain (HK) Educational Academy, Software House & Training Center\contact managment system>
```

Data Save Successfully in JSON File
Updated Data:

Edit Selection View Go Run ··· contact managment system

XPLORER ··· main.py | contacts.json | requirements.txt | README.md

ONTACT MANAGMENT S...
- contacts.json
- main.py
- README.md
- requirements.txt

contacts.json > { } 2


```
[
    {
        "id": 2,
        "name": "Ahmed",
        "phone": "03332332445",
        "email": "ahmed@gmail.com",
        "city": "islamabad",
        "company": "abc software",
        "favorite": false,
        "created_at": "2026-04-28 18:14:51.087983"
    },
    {
        "id": 1,
        "name": "Hashir Ahmed Buriro",
        "phone": "+923332332600",
        "email": "buriro.hashir@gmail.com",
        "city": "Karachi",
        "company": "UQ Consultants",
        "favorite": true,
        "created_at": "2026-04-28 10:00:00"
    },
    {
        "id": 4,
        "name": "Hashir Buriro",
        "phone": "03332332600",
        "email": "hashir@gmail.com",
        "city": "karachi",
        "company": "UQ Consltants",
        "favorite": true,
        "created_at": "2026-04-28 18:16:06.683059"
    }
]
```


UTLINE

MELINE

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Result

The developed Contact Management System was fully tested and performed as expected. All functional requirements were successfully implemented, including contact creation, advanced search, filtering, updates, deletion, and data persistence using JSON file handling. The system provides a smooth and user-friendly CLI experience with proper validation and error handling. Overall, the project meets the internship requirements and demonstrates strong understanding of Python programming fundamentals and real-world application development.

Conclusion

This Contact Management System project provides a practical implementation of a mini CRM built in Python. It focuses on real-world functionality such as contact storage, advanced searching, filtering, and data management using a JSON-based persistence system. The project demonstrates strong understanding of modular programming, file handling, input validation, and CLI-based user interaction. It serves as a solid foundation for building more advanced systems like database-driven CRM or GUI applications in the future.